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Altered gravity and time perception: a brief review

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Gravity is fundamental to life on Earth, shaping biological systems and perceptual expectations. Understanding how gravity influences cognition is critical as humans increasingly encounter altered-gravity environments through spaceflight and parabolic flights. This mini-review synthesizes emerging findings on how deviations from Earth-normal gravity affect human time perception. Studies across fields of psychology, neuroscience, and physiology suggest that time perception reflects the dynamic integration of multisensory and bodily signals anchored to Earth's gravitational constraints. We identify key gaps in linking physiological changes to behavioral performance and in extending research to longer time scales, areas needing further advancement to benefit space travelers and offer insights into mechanisms of human timing.

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Introduction

Gravity is an ever-present force that has shaped the evolution of physiological and cognitive processes (Adamopoulos et al., 2021; Morey-Holton, 2003). However, humans increasingly encounter environments where gravity conditions differ from Earth's through parabolic flights and space travel, as well as analogous experiences in virtual or augmented reality (VR/AR). These scenarios provide novel interactions with altered gravitational cues, motivating a recent surge in scientific research on the perceptual and biological effects of such alterations. Existing reviews have focused on how Earth's gravity influences aspects of perception and action, such as spatial orientation and motion (Dakin and Rosenberg, 2018; Jörges and López-Moliner, 2017; Lacquaniti et al., 2015; Zago and Lacquaniti, 2005). This review considers recent studies on how Earth-discrepant gravity influences time perception.

Accurate timing is crucial for predicting and coordinating movements in response to environmental stimuli (Breska and Ivry, 2018; Coull and Nobre, 2008; Herbst et al., 2022). Time perception may be distorted under various internal or external conditions, requiring adaptive strategies to achieve task goals. This is of practical consideration for aerospace training and preparation, as astronauts undertake complex tasks in altered-gravity environments. Despite the ubiquity and relevance of time as a feature of human behavior and cognition, much remains unknown about the psychological and neurological mechanisms of time perception, and even less is known about how these processes are affected by gravity. Understanding gravitational effects on time perception may provide insight into the underlying mechanisms of human timing under normal Earth gravity (1 g) conditions, as well as improving the health and performance of astronauts during and after space travel.

The goal of this preregistered review is to clarify patterns of results from empirical studies examining time perception in differing gravitational contexts and how physiological changes may mediate these perceptual effects to identify knowledge gaps and potential targets for future research. In the first section, we focus on recent evidence for gravitational effects on time perception across a variety of timing tasks. We consider the effects of altered gravitational cues on time perception contrasted against the temporal referent of persistent gravitational forces on Earth. Second, we focus on how transient or persistent physiological changes may be linked to altered time perception by reviewing relations between gravitational forces and neural and physiological processes relevant to timing and time perception. Finally, we discuss open questions and future directions for research.

Altered gravity affects time perception

Gravity affects our perception of time in two key ways: indirectly by influencing the motion of external objects in the world, or directly through changes in physiological or sensory inputs. A substantial body of evidence links the perception of visual object motion through space to time perception (e.g., Brown, 1995; Conway et al., 2016; He et al., 2023; Huang and Jones, 1982; Vidaud-Laperrière et al., 2022), with fewer studies considering the mediating role of gravity on this space-time relation (for a review see: Lacquaniti et al., 2015). From a Bayesian estimation framework, human development and evolution within Earth's gravity provides a strong prior for expected motion physics, and deviations from this prior may induce temporal errors (for review; Jörges and López-Moliner, 2017). Changes in gravitational forces, which are typically stable when at rest on Earth, can also cause physiological symptoms, such as those related to cephalic fluid shifts, affecting the optic and vestibular systems (Hallgren et al., 2016; Lawson et al., 2016; Mader et al., 2013; Norsk et al., 2015). Thus, studying how gravitational forces affect time perception can involve experimentally manipulating the motion kinematics of objects in the visual field or by altering vestibular or proprioceptive feedback. The studies reviewed in this section used various approaches to test time perception under different experimental gravitational environments, including simulated and real gravitational changes experienced by astronauts on the International Space Station (ISS) or during parabolic flight.

Simulated gravity changes

Visual motion kinematics

Gravity affects how objects in the natural world move. Human observers primarily experience visual events in an upright orientation, whereby objects in free-fall accelerate downward. Both the orientation and acceleration of motion have been implicated in timing judgments. Objects moving downward in the visual field produce more accurate duration judgments than objects moving in an orthogonal plane (Moscatelli and Lacquaniti, 2011; Torok et al., 2019). Moscatelli and Lacquaniti (2011) found that participants more accurately judged the duration of an object

moving a fixed distance with constant acceleration (9.81 m/s^2 —approximate free-fall acceleration in Earth's gravity) when it was moving downward (congruent with gravity) than upward or sideways. This effect was present even when participants were lying on their side, suggesting that they incorporate proprioceptive gravitational information in addition to visual orientation for duration perception. In a VR study, Torok et al. (2019) found a similar gravitational bias for objects moving a fixed distance under normal and reduced acceleration conditions that simulated Earth and Mars gravity, further supporting the downward orientation-based bias. These studies indicate that, while the visual system may flexibly account for factors affecting acceleration, such as air resistance, duration perception is best for objects moving in a gravity-congruent direction.

Other evidence suggests that gravitational acceleration may be more relevant for the timing of self-motion. One study manipulated visual self-motion on a simulated rollercoaster while participants performed temporal predictions (Indovina et al., 2013). Conditions included downward acceleration or upward deceleration consistent with gravity, and horizontal motion with matched acceleration and deceleration motion parameters (null gravity conditions). Results showed that acceleration and orientation affected the temporal prediction of self-motion. Participants produced anticipatory (early) prediction responses under acceleration and delayed (late) responses under deceleration compared to a constant velocity control condition, and earlier temporal anticipation for downward compared to horizontal acceleration.

Another type of self-motion timing affected by visual kinematics is the temporal control of limb movements during synchronization with a steady (i.e., isochronous) visual rhythm. Previous work has shown that participants exhibit better synchronization performance when tapping a finger with a visual metronome featuring continuous-motion properties (i.e., a bouncing ball) compared to discrete flashes (i.e., a blinking light; Iversen et al., 2015). One study by Gan et al. (2015) showed improved finger-tapping synchronization to a bouncing ball with a realistic acceleration profile matched to Earth's gravity compared to less “natural” motion lacking gravitational acceleration. Thus, gravity-congruent motion enhanced temporal prediction, increasing precision in repetitively timed movements. Another study showed better synchronization with accelerating compared to decelerating motion kinematics, but no orientation effects (Pérez et al., 2023). Overall, altering visual gravitational acceleration and orientation affects temporal prediction during passive and active self-motion tasks.

Physiological cues

The simulation studies described above lack physiologically-coherent changes under altered-gravity conditions, producing a mismatch in perceptual and physiological cues. While the visual system is experiencing altered gravity, the observer is still subject to Earth's gravitational inputs to vestibular, proprioceptive, and cardiovascular systems, which, as discussed later in this review, have relevant influences on time perception. Head-down bed rest (HDBR) posture has been used since the 1970s to simulate microgravity (0 g) effects on the body, requiring participants to

lie in a -6° inversion for a prolonged period of time (Traon et al., 2007). Vestibular effects of HDBR (Yuan et al., 2018) affect magnitude and time estimation (Gammeri et al., 2023; Qian et al., 2021). Qian et al. (2021) approximated cephalic fluid shifts and vestibular changes experienced in 0 g through an HDBR protocol. Participants performed duration judgments on the 8th and 15th days of consistent HDBR as well as 1 week before and after. Temporal sensitivity thresholds increased from baseline across the HDBR period, returning to baseline 1 week post-HDBR, indicating that physiological changes associated with microgravity contribute to worse time perception. Reduced vestibular inputs due to bilateral vestibulopathy (BVP) share similarities with microgravity effects, wherein fluid-filled peripheral vestibular structures lack reliable information from Earth's gravito-inertial forces during head tilt. Clément et al. (2023) compared astronauts' duration perception during spaceflight to that of BVP patients. Astronauts tended to underestimate a 1-min duration in spaceflight compared to on Earth, demonstrating similar duration perception in 0 g as BVP subjects in 1 g.

Real gravity changes

Gravitational effects have been observed for duration perception and reaction time for humans experiencing weightlessness (i.e., 0 g) during parabolic flight or spaceflight. Clément (2018) evaluated supra-second duration estimation during repeated bouts of 1, 0, and 1.8 g lasting 25–30 s each during parabolic flight sequences. Passengers were required to estimate the duration of a visual marker while performing a digit-reading task to prevent explicit counting of seconds. Durations were underestimated in 0 and 1.8 g relative to 1 g, however, only 0 g conditions produced statistically-significant underestimations. Navarro-Morales et al. (2023) found a similar result for astronauts, estimating 1-min by counting before, during, and after spending 6–8 months on the ISS. Compared to pre- and post-flight estimations, 1-min was underestimated when free-floating on board the ISS. On average, astronauts in 0 g reported that a minute had passed after a duration 15-s shorter compared to normal 1 g. Another study reported similar temporal underestimation while aboard the ISS compared to pre- and post-flight with durations ranging from 2 to 38 s (Kuldavletova et al., 2023). This study additionally showed a trend of increasing temporal error (greater underestimation) across days 17–164 on the ISS with repeated testing for a dual-task requiring simultaneous digit-reading. For reaction time tasks, participants were slower to press a button following a visual cue in 0 g compared to 1 or 1.8 g during parabolic flight (Clément, 2018) and spaceflight (Kuldavletova et al., 2023). Together, results suggest that time “passes faster” in altered gravity, and increased cognitive load may play a role in duration underestimation and slower reaction times produced by astronauts in spaceflight.

Neurophysiological mediation of gravitational effects on timing

Of utmost consideration for understanding gravitational effects on time perception are potential neurological and physiological

mediators. Increases and decreases in gravitational forces on the body cause physiological changes to cardiovascular, vestibular, ocular, and somatosensory systems (Lee et al., 2020; Mergner and Rosemeier, 1998; Reschke and Clément, 2018b; Sarker et al., 2023). In astronauts exposed to different gravity conditions, changes include reduced heart rate and heart rate variability (Fritsch-Yelle et al., 1985; Migeotte et al., 2003; Otsuka et al., 2016), increased saccade latency (Reschke et al., 2017), and altered pupil activity associated with increased intracranial pressure and spaceflight-associated neuro-ocular syndrome (Lee et al., 2020; Sarker et al., 2023). Such bodily functions have been associated with timing processes; thus, altered gravity may affect time perception mechanisms directly or indirectly through neurophysiological mediators.

Neurophysiological correlates of time perception in Earth's gravity

Time perception has been linked to physiological metrics such as temperature (Van Maanen et al., 2019), heart rate (Pollatos et al., 2014; Arslanova et al., 2023), pupil dilation (Faber, 2017), and eye movements (Ayres et al., 2021). Greater heart rate variability is associated with more accurate reproduction of temporal intervals, especially for intervals between 2 and 25 s (Pollatos et al., 2014). Furthermore, increased pupil size correlates with higher cognitive load and error monitoring during timing tasks (Eckstein et al., 2017; Warda et al., 2022). Changes in arousal also affect time perception, where higher arousal, marked by increased heart rate, body temperature, and pupil diameter, is associated with faster tempo judgments and duration overestimation (Jakubowski et al., 2015; Wearden and Penton-Voak, 1995).

The autonomic nervous system plays a central role in these effects. According to interoceptive theories (Craig, 2002, 2009), time perception arises from signals sent by afferent autonomic pathways conveying the internal states. Popular models of timing suggest that an internal pacemaker's rate and attentional focus on time passage govern subjective timing (Zakay and Block, 1997; Brown, 2008). Complementary neurophysiological evidence indicated that subjective time perception engages partially distinct neural networks depending on sensory context. Proshina et al. (2024) found that duration reproduction under eyes-open and eyes-closed conditions elicited differential activation patterns across parietal, frontal, and cingulate regions using EEG source localization. While subjective temporal judgments did not vary behaviorally across conditions, oscillatory activity in beta and alpha bands correlated with duration estimates within domain-relevant cortical sites, suggesting that context-dependent recruitment of cortical resources may reflect flexible instantiations of time perception processes. Pollatos et al. (2014) showed that sympathovagal balance and interoceptive processing contribute to temporal accuracy, indicating that cardiac signals play a direct role in interval timing. They found that individuals with higher interoceptive sensitivity demonstrated better synchronization performance specifically for a 2-s interval, suggesting that the influence of bodily signals on timing may depend on the duration being judged. Lacquaniti et al. (2015) proposed that the brain

incorporates an internal model of gravitational effects with multisensory signals for temporal and spatial tasks, underscoring the integration of multimodal gravity-related information, such as visual object motion and vestibular inputs, for time perception.

Gravitational effects on neurophysiology

Spaceflight and artificial gravity studies reveal that altered gravitational loads modulate physiological rhythms interconnected with timing. Microgravity reduces parasympathetic tone and heart rate variability, altering autonomic cardiovascular control and potentially destabilizing interoceptive signals important for timing (Fritsch-Yelle et al., 1985; Migeotte et al., 2003; Otsuka et al., 2016). These physiological parameters often persist upon initial return to Earth's gravity and later reverse, with elevated heart rates commonly observed within 24 h post-flight and persisting for days or weeks before returning to baseline. Simultaneously, vestibular perturbations modulate spatial and temporal perception through parietal-insular cortical pathways sensitive to gravitational cues (Denise et al., 2022; Walsh, 2003). The interaction of gravitational cues with visual and vestibular mechanisms further underscores the sensorimotor grounding of temporal representations. Visual-motor coordination and oculomotor dynamics are also influenced by gravity. For example, pupil and saccadic alterations tied to intracranial pressure changes impact timing-related cognitive load and temporal integration during spaceflight (Lee et al., 2020; Sarker et al., 2023; Reschke and Clément, 2018a). Altogether, gravitational modulation of physiological processes, including autonomic cardiac signals, vestibular input, ocular-motor function, and arousal states, forms a complex interplay that affects the experience of time. Understanding these neuro-psychophysiological signatures offers insights into time perception for future space missions (Craig, 2002, 2009; Fritsch-Yelle et al., 1985; Lacquaniti et al., 2013).

Taken together, these lines of research support the notion that the perception of time is subserved by a complex, distributed network including sensory, vestibular, and parietal-insular cortical regions, with some task-specific engagement. Crucially, vestibular and gravitational inputs appear to modulate these mechanisms, and conditions associated with normal Earth gravity provide a fundamental reference for temporal estimation. Emerging evidence from altered gravity environments, neuroimaging, and electrophysiology suggests that temporal processing is not only symbolically abstract but also deeply tied to bodily states and ecological constraints. Thus, these studies provide an indirect link between gravity-based neurophysiological changes to altered time perception.

Future directions and conclusion

Much remains unknown regarding the effects of altered gravity on time perception via psychological or neurophysiological changes. Although there is a growing literature on gravitational effects on human physiological and perceptual functions, most studies have examined these effects in isolation, requiring causal

inferences between relevant changes. Future studies should directly examine the roles of vestibular, visual, and interoceptive systems in time perception under altered gravity. Multimodal neuroimaging and electrophysiological studies during space missions and analog environments could clarify the underlying neural and physiological mechanisms of timing. Longitudinal monitoring of astronauts before, during, and after flight is essential to characterize adaptive changes in temporal processing and recovery. Including cross-modal timing tasks spanning visual, auditory, and tactile domains will help reveal how gravity-related sensory alterations shape integrated time perception. Further, ecologically valid tasks that mimic activities undertaken by astronauts on spaceflight missions will increase the translational relevance of findings.

Continued development and testing of countermeasures such as immersive VR or physiological feedback protocols will be important from both basic and applied scientific perspectives. Considering studies under real gravitational manipulations, isolating sensory and physiological effects using simulations may help disentangle contributions of different sources of information such as visual dynamics and vestibular inputs. Altered gravity not only causes astronauts to feel the effects of changes in vestibular inputs but also fundamentally changes the nature of how objects move around the visual field. VR/AR studies provide an avenue for testing the effects of visual gravitational changes on time perception in the absence of vestibular alteration. Understanding the connection between gravity-based changes and visual information is crucial not only because of physiological impacts on the visual system (Lee et al., 2020; Reschke et al., 2017; Sarker et al., 2023), but because visual object motion directly modulates time perception for stimuli within and across modalities (Brown, 1995; He et al., 2023).

Future studies should go beyond vestibular and proprioceptive to consider auditory, tactile, and interoceptive signals in gravity-modulated timing. Multisensory tasks can distinguish modality-specific from shared effects. How gravity interacts with affective and cognitive states, such as stress or motivation, also warrants attention. Active timing (timed motor actions) should be contrasted with passive timing (perception without action), as altered gravity may affect motor-perceptual integration differently from perception alone. With growing potential for space missions to extend in duration and distance, the systematic investigation of longer timescales should be undertaken. Most prior work has focused on brief (milliseconds-seconds) or intermediate (seconds-minutes) intervals, leaving hours-to-days durations relatively unexplored. Examining circadian (24-h) and ultradian (less than 24-h) rhythms in altered gravity will further clarify mechanisms governing timing across scales.

Finally, advancing formal computational models should explicitly incorporate gravity as a parameter in timing mechanisms. Such models should account for dynamic re-weighting of multisensory inputs, particularly vestibular, visual, proprioceptive, and interoceptive signals, when reliability changes under different gravitational conditions. Bayesian perceptual inference (Jörges and López-Moliner, 2017, 2020; Allred et al., 2023), Bayesian Prediction Error Minimization (PEM; i.e., “distrusting the present”; Kent et al., 2019), and predictive-processing frameworks (Millidge et al., 2024) offer promising foundations as they can simulate how altered

gravity introduces temporal biases by modifying prediction errors and sensory precision estimates. State-space adaptation models (e.g., Ravichandran-Schmidt and Hass, 2024) may further help chart how perceptual timing recalibrates over longer missions and, once astronauts return to Earth, predict trajectories of learning, compensation, and recovery.

In conclusion, growing evidence highlights that altered gravity profoundly influences perceptual, neural, and physiological substrates of human timing. These changes manifest as distortions in subjective timing, variability in temporal accuracy, and recalibration of sensorimotor integration. Understanding how gravity reshapes time perception not only informs fundamental cognitive neuroscience but is critical for astronaut performance and wellbeing during long-duration space missions. Continued interdisciplinary research integrating psychology, neurophysiology, vestibular science, and medicine is vital to unraveling these complex processes and to developing effective countermeasures for the challenges of human activities beyond Earth. Such studies may also extend beyond space exploration, offering translational benefits for clinical neuroscience. Simulations such as VR/AR can be harnessed for rehabilitation protocols that allow patients with motor, vestibular, or perceptual disorders to relearn and practice timing-dependent skills in controlled, adaptive environments. In this way, research on gravity and time perception not only advances our readiness for future spaceflight but also enriches therapeutic strategies to support recovery and functional independence on Earth.

Author contributions

AJ-N: Writing – original draft, Conceptualization, Writing – review & editing, Methodology. CK: Methodology, Writing – review & editing, Conceptualization, Writing – original draft.

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